```
mysql> SELECT * FROM MOVIE ORDER BY movie_id;
+----------+-------------------------+------+------------------+
| movie_id | title                   | year | director         |
+----------+-------------------------+------+------------------+
|      101 | Gone with the Wind      | 1939 | Victor Fleming   |
|      102 | Star Wars               | 1977 | George Lucas     |
|      103 | The Sound of Music      | 1965 | Robert Wise      |
|      104 | E.T.                    | 1982 | Steven Spielberg |
|      105 | Titanic                 | 1997 | James Cameron    |
|      106 | Snow White              | 1937 | NULL             |
|      107 | Avatar                  | 2009 | James Cameron    |
|      108 | Raiders of the Lost Ark | 1981 | Steven Spielberg |
+----------+-------------------------+------+------------------+
```

```
mysql> SELECT * FROM REVIEWER ORDER BY reviewer_id;
+-------------+------------------+
| reviewer_id | reviewer_name    |
+-------------+------------------+
|         201 | Sarah Martinez   |
|         202 | Daniel Lewis     |
|         203 | Brittany Harris  |
|         204 | Mike Anderson    |
|         205 | Chris Jackson    |
|         206 | Elizabeth Thomas |
|         207 | James Cameron    |
|         208 | Ashley White     |
+-------------+------------------+
```

```
mysql> SELECT * FROM RATING ORDER BY reviewer_id, movie_id, rating_date;
+-------------+----------+-------+-------------+
| reviewer_id | movie_id | stars | rating_date |
+-------------+----------+-------+-------------+
|         201 |      101 |     2 | 2011-01-22  |
|         202 |      106 |     4 | NULL        |
|         203 |      103 |     2 | 2011-01-20  |
|         203 |      108 |     4 | 2011-01-12  |
|         204 |      101 |     3 | 2011-01-09  |
|         205 |      103 |     3 | 2011-01-27  |
|         205 |      104 |     2 | 2011-01-22  |
|         205 |      108 |     4 | NULL        |
|         206 |      106 |     5 | 2011-01-19  |
|         206 |      107 |     3 | 2011-01-15  |
|         207 |      107 |     5 | 2011-01-20  |
|         208 |      104 |     3 | 2011-01-02  |
+-------------+----------+-------+-------------+
```

Task 1

```
>> SELECT
      movie.movie_id,movie.title,rating.stars
   FROM
       movie
   LEFT JOIN
       rating on movie.movie_id = rating.movie_id;
```

movie_id	title	stars
101	Gone with the Wind	2
101	Gone with the Wind	3
102	Star Wars	NULL
103	The Sound of Music	2
103	The Sound of Music	3
104	E.T.	2
104	E.T.	3
105	Titanic	NULL
106	Snow White	4
106	Snow White	5
107	Avatar	3
107	Avatar	5
108	Raiders of the Lost Ark	4
108	Raiders of the Lost Ark	4

```
>> SELECT
      movie.movie_id,movie.title,rating.stars
   FROM
      movie
   JOIN
      rating on movie.movie_id = rating.movie_id;
```

movie_id	title	stars
101	Gone with the Wind	2
101	Gone with the Wind	3
103	The Sound of Music	2
103	The Sound of Music	3
104	E.T.	2
104	E.T.	3
106	Snow White	4
106	Snow White	5
107	Avatar	3
107	Avatar	5
108	Raiders of the Lost Ark	4
108	Raiders of the Lost Ark	4

```
>>SELECT
        movie.movie_id,movie.title,reviewer.reviewer_id,rating.stars
    FROM
        movie
    LEFT JOIN
        rating ON movie.movie_id = rating.movie_id
    LEFT JOIN
        reviewer ON rating.reviewer_id = reviewer.reviewer_id;
```

movie_id	title	reviewer_id	stars
101	Gone with the Wind	201	2
101	Gone with the Wind	204	3
102	Star Wars	NULL	NULL
103	The Sound of Music	203	2
103	The Sound of Music	205	3
104	E.T.	205	2
104	E.T.	208	3
105	Titanic	NULL	NULL
106	Snow White	202	4
106	Snow White	206	5
107	Avatar	206	3
107	Avatar	207	5
108	Raiders of the Lost Ark	203	4
108	Raiders of the Lost Ark	205	4

```
>>SELECT
      movie.movie_id,movie.title,reviewer.reviewer_id,rating.stars
    FROM
      movie
    JOIN
      rating ON movie.movie_id = rating.movie_id
    JOIN
      reviewer ON rating.reviewer_id = reviewer.reviewer_id;
```

movie_id	title	reviewer_id	stars
101	Gone with the Wind	201	2
101	Gone with the Wind	204	3
103	The Sound of Music	203	2
103	The Sound of Music	205	3
104	E.T.	205	2
104	E.T.	208	3
106	Snow White	202	4
106	Snow White	206	5
107	Avatar	206	3
107	Avatar	207	5
108	Raiders of the Lost Ark	203	4
108	Raiders of the Lost Ark	205	4

TASK 2

Select movie ids of each movie with its title, reviewer id and stars received, where number of stars are less than or equal to three.

```
>>  SELECT
        movie.movie_id,movie.title,reviewer.reviewer_id,rating.stars
    FROM
        movie
    JOIN
        rating ON movie.movie_id = rating.movie_id
    JOIN
        reviewer ON rating.reviewer_id = reviewer.reviewer_id
    WHERE
        rating.stars<=3;
```

movie_id	title	reviewer_id	stars
101	Gone with the Wind	201	2
103	The Sound of Music	203	2
101	Gone with the Wind	204	3
103	The Sound of Music	205	3
104	E.T.	205	2
107	Avatar	206	3
104	E.T.	208	3

7 rows in set (0.00 sec)

TASK 3

Select each movie id with its corresponding title, reviewer id, reviewer name and stars received.

```
>> SELECT
       movie.movie_id,movie.title,reviewer.reviewer_id,reviewer.reviewe
r_name,rating.stars
   FROM
       movie
   JOIN
        rating ON movie.movie_id = rating.movie_id
   JOIN
        reviewer ON rating.reviewer_id = reviewer.reviewer_id;
```

TASK 4

Select each movie id with its corresponding title, reviewer id, reviewer name and stars received, where the number of stars received is equal to five

```
>> SELECT
        movie.movie_id,movie.title,reviewer.reviewer_id,reviewer.reviewe
r_name,rating.stars
    FROM
        movie
    JOIN
        rating ON movie.movie_id = rating.movie_id
    JOIN
        reviewer ON rating.reviewer_id = reviewer.reviewer_id
    WHERE
        rating.stars=5;
```

TASK 5

Select movie title with its corresponding reviewer name and stars, where the movie's rating date is missing.

```
>> SELECT
         movie.title,reviewer.reviewer_name,rating.stars
    FROM
        movie
    JOIN
        rating ON movie.movie_id=rating.movie_id
    JOIN
        reviewer ON rating.reviewer_id = reviewer.reviewer_id
    WHERE
        rating.rating_date IS NULL;
```

TASK 6

Select the details about the reviewers who have a last name called 'Martinez'.

```
>>  SELECT *
    FROM reviewer
    WHERE reviewer_name LIKE '%Martinez';
```

TASK 7

Show the effect of giving one more star to the movies reviewed by 'Brittany Harris'. Here select relevant details from the RATING table.

```
>>SELECT
      rating.movie_id,
      rating.reviewer_id,
      rating.stars AS original_stars,
      rating.stars+1 AS increased_stars
   FROM
      rating
   JOIN
      reviewer ON rating.reviewer_id = reviewer.reviewer_id
   WHERE
      reviewer.reviewer_name='Brittany Harris';

>> SELECT
     movie_id,
     reviewer_id,
     stars AS original_stars,
     stars+1 AS increased_stars
   FROM
     rating
   WHERE
       reviewer_id=(SELECT reviewer_id FROM reviewer WHERE
       reviewer_name='Brittany Harris');
```

Task 8

Select movie titles with its reviewer name and stars received. Order the result by movie title in the alphabetical order.

```
>>SELECT
      movie.title,
      reviewer.reviewer_name,
      rating.stars
   FROM
      movie
   JOIN
      rating ON movie.movie_id=rating.movie_id
   JOIN
     reviewer ON rating.reviewer_id=reviewer.reviewer_id
   ORDER BY
     movie.title ASC;
```

TASK 9

Write a nested query to list the details of the movies directed by a director A. who is also a reviewer. B. who is not a reviewer.

```
>> SELECT *
     FROM
   movie
     WHERE
      movie.director IN (SELECT reviewer_name FROM reviewer);
```

TASK 10

Write a nested query to list the details of the movie ratings, A. reviewed by the reviewer ‘Sarah Martinez’.

```
>> SELECT *
   FROM
     rating
   WHERE
     reviewer_id=(SELECT reviewer_id FROM reviewer WHERE
     reviewer_name='Sarah Martinez');
```

TASK 11

Find all the years that have a movie that received a rating of 4 or 5 and sort them in increasing order of the year. Write,
A. a non-nested query.

```
>>   SELECT
       movie.year
     FROM
       movie
     JOIN
       rating ON movie.movie_id=rating.movie_id
     WHERE
       rating.stars = 4 OR rating.stars = 5
     ORDER BY
       movie.year ASC;
```

```
    ->  mov
+------+
| year |
+------+
| 1937 |
| 1937 |
| 1981 |
| 1981 |
| 2009 |
+------+
```

```
>> SELECT
    DISTINCT movie.year
   FROM
      movie
   JOIN
      rating ON movie.movie_id=rating.movie_id
   WHERE
      rating.stars = 4 OR rating.stars = 5
   ORDER BY
      movie.year ASC;
```

```
+------+
| year |
+------+
| 1937 |
| 1981 |
| 2009 |
+------+
```

B. a nested query.

```
>> SELECT
    DISTINCT movie.year
   FROM
      movie
   WHERE
      movie_id IN (SELECT movie_id FROM rating WHERE stars=4 or
stars=5);
```

TASK 12

List the movie titles and average ratings, from the highest-rated to lowest-rated. If two or more movies have the same average rating, list them in alphabetical order.

```
>> SELECT
      movie.title,
      AVG(rating.stars) AS average_rating
   FROM
      movie
   JOIN
      rating ON movie.movie_id = rating.movie_id
   GROUP BY
      movie.title
   ORDER BY
      average_rating DESC,
      movie.title ASC;
```

TASK 13

Remove all ratings where the movie's year is before 1970 or after 2000

```
>> DELETE FROM
       rating
   WHERE
       movie_id IN (SELECT movie_id FROM movie WHERE year<1970 OR
year > 2000 );
```